# Results and Discussion

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## 1 Research Question

Reddit User Churn Prediction Study: do Reddit users show detectable behavioral shifts in language use and social network activity before they disengage (RQ1), and does combining NLP-based features with social network features produce meaningfully better churn predictions than either feature type used alone (RQ2)?

## 2 Initial Results

### 2.1 Exploratory Data Analysis

#### 2.1.1 Data

Data were collected from three subreddits via Arctic Shift, covering each subreddit's available history from creation through January 2026: r/learnprogramming, r/loseit, and r/depression. For each subreddit, both posts and comments were downloaded, including author, timestamp, text content, and parent ID.

Cleaning was applied consistently across all three subreddits: duplicate records were removed, accounts marked as [deleted] or [removed] were excluded, known bots such as AutoModerator were filtered out, and users in the bottom 5 percent of activity within each subreddit were dropped. After cleaning, r/learnprogramming contains 517,079 posts and 4,017,272 comments from 601,498 qualifying users; r/loseit contains 521,920 posts and 7,725,732 comments from 792,129 qualifying users; and r/depression contains 1,330,924 posts and 5,323,107 comments from 1,234,157 qualifying users.

### 2.1.2 User Activity Distribution

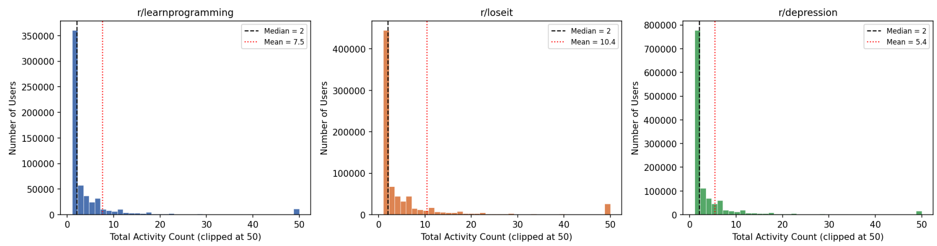


Figure 1: User Activity Distribution Across Three Subreddits

From Figure 1, the activity distribution is strongly right-skewed in all three subreddits. The median user has only 2 total activities across all three communities, while the mean is much higher, ranging from 5.4 in r/depression to 10.4 in r/loseit. This shows that most users participate only a few times, while a small group of highly active users contributes much more content, which is consistent with prior research on large online communities (Chandrasekharan et al., 2022). There are also differences across communities. r/loseit has the highest mean activity at 10.4, which may reflect the fact that weight loss is a process that unfolds over months and users may return repeatedly to share progress updates. r/learnprogramming sits in the middle, with a mean of 7.5, suggesting that many users may come with specific questions and return only when they encounter new problems, while r/depression has the lowest mean activity at 5.4, suggesting that participation in this community may be more brief and episodic.

### 2.1.3 Calibration Period Activity: Frequency and Recency

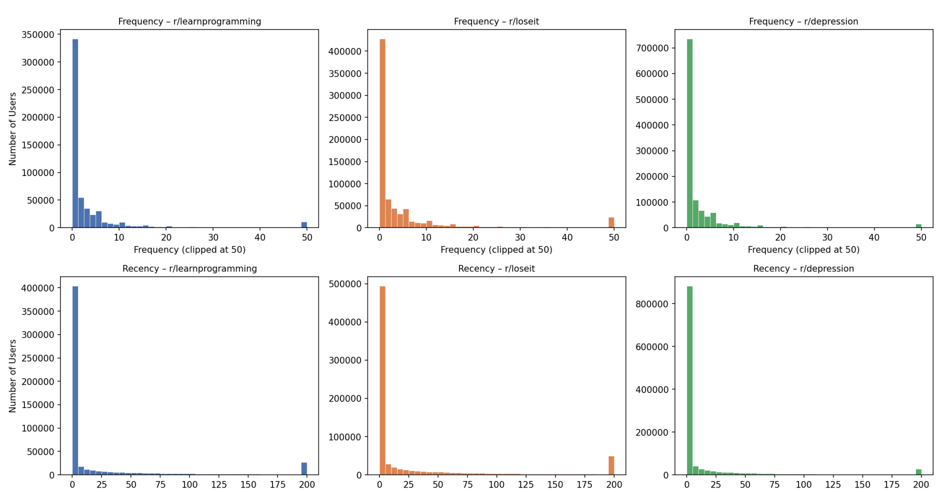


Figure 2: Frequency (top row) and Recency (bottom row) Distributions

From Figure 2, both frequency and recency are strongly right-skewed across all three subreddits. Most users contributed only a small number of posts or comments during the calibration period. The recency distributions also show a sharp spike at zero weeks, indicating that many users had their first and last observed activity within the same week. This suggests that participation is often short-term rather than sustained over time. The recency pattern also differs across communities. r/loseit has a more spread-out distribution, suggesting more repeated participation, while r/depression has the strongest spike at zero weeks, suggesting more brief and episodic participation. This pattern is consistent with the activity differences shown in Figure 1.

These frequency and recency distributions are directly relevant to the churn labeling approach. The BG/NBD framework, which models posting behavior as a probabilistic process with individual-level dropout probabilities, is well-suited to irregular, right-skewed activity patterns like those observed here (Fader et al., 2005). However, as described in Section 3.2, the model failed to converge in this dataset, and churn was ultimately defined using a holdout window instead.

### 2.1.4 Reply Network Structure

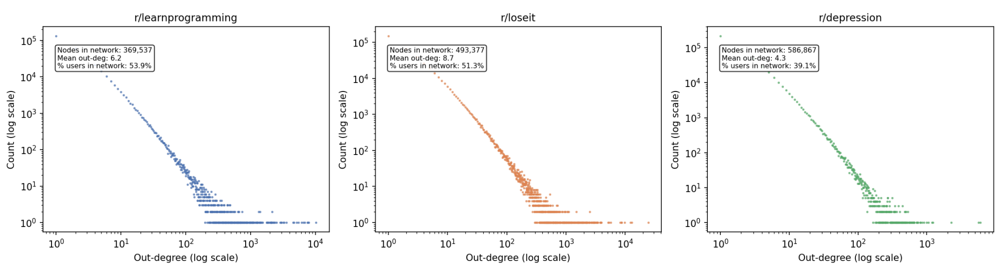


Figure 3: Reply Network Out-Degree Distribution Across Three Subreddits

From Figure 3, the out-degree distributions follow an approximate power-law pattern in all three subreddits, with a small number of users sending a very large number of replies and the vast majority having out-degree of one or two. However, the three communities differ in important ways. The mean out-degree is highest in r/loseit, at approximately 8.7, and lowest in r/depression, at approximately 4.3. This may reflect the more conversational and supportive exchange culture in the weight loss community, compared with the more isolated posting pattern in the mental health community.

The notably lower network engagement in r/depression is especially important for this study. A majority of r/depression users, about 60.9 percent, never replied to another user during the observation period, compared with roughly half of users in the other two communities. This suggests that for many r/depression users, the community may function more as a one-way outlet for self-expression. This structural difference has an important implication for the modeling stage: network features may capture less behavioral information for r/depression users, and the relative predictive value of NLP features and network features may differ across communities.

## 2.2 Initial Analysis

To address RQ1, the analysis is divided into two parts. The first part compares static feature differences between churned and active users, including both NLP and network metrics (Figures 4 and 5). The second part examines how user activity changes in the months before disengagement (Figure 6).

### 2.2.1 Static Feature Differences

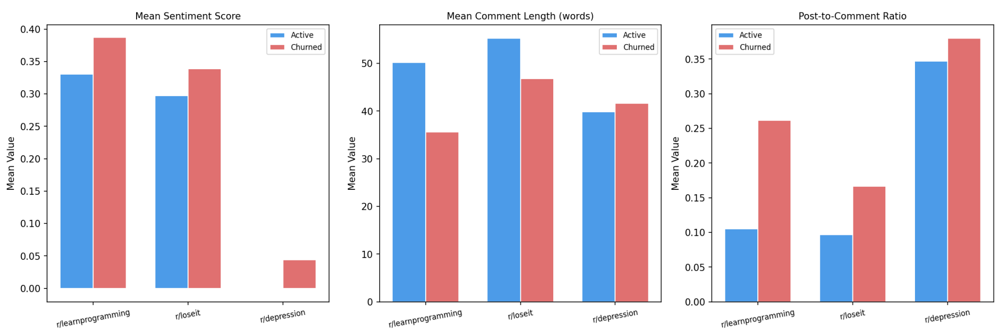


Figure 4: NLP feature differences across three subreddits.

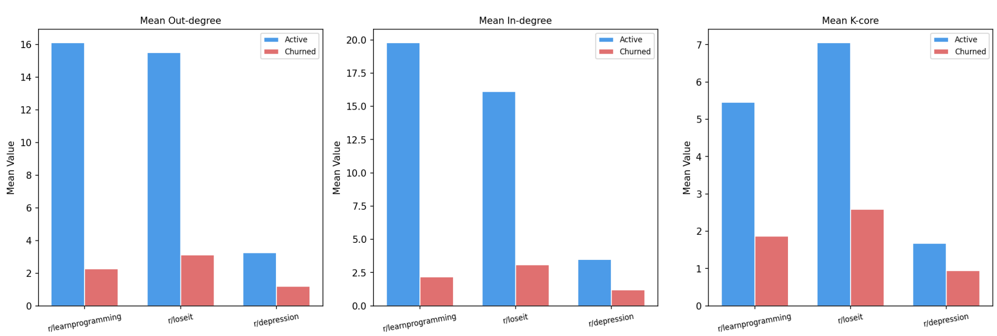


Figure 5: Network feature differences across three subreddits.

Firstly, NLP features show some differences between churned and active users, but the differences are generally small. The clearest pattern is that churned users tend to write shorter comments and have higher post-to-comment ratios. Churned users also had higher post-to-comment ratios, such as 0.26 versus 0.10 in r/learnprogramming and 0.17 versus 0.10 in r/loseit. This suggests that churned users are more likely to post directly than to participate in longer reply-based conversations. However, sentiment differences are small and unstable, so sentiment alone does not appear to be a strong churn signal. This result fits with prior churn prediction research showing that textual features can be useful, but they are often stronger when combined with behavioral or relational features rather than used alone (Shahabikargar et al., 2026; Klugman, 2023).

Secondly, network features show much clearer differences than NLP features. Active users have higher out-degree, in-degree, and k-core values across all three subreddits. In r/learnprogramming, active users had a mean out-degree of 16.1, compared with 2.3 for churned users. Their mean in-degree was also much higher, 19.8 compared with 2.2. In r/loseit, active users had a mean out-degree of 15.5, compared with 3.1 for churned users. These gaps suggest that active users are more embedded in the reply network, while churned users remain more peripheral.

However, this network-based pattern is weaker in r/depression. Active users in r/depression had a mean out-degree of only 3.2, compared with 16.1 in r/learnprogramming and 15.5 in r/loseit. Churned users in r/depression had a mean out-degree of only 1.2. This suggests that r/depression has a weaker reply-based structure, and many users may participate without forming repeated interactions. As a result, network features may be less informative in r/depression than in the other two communities.

### 2.2.2 Dynamic Feature Differences

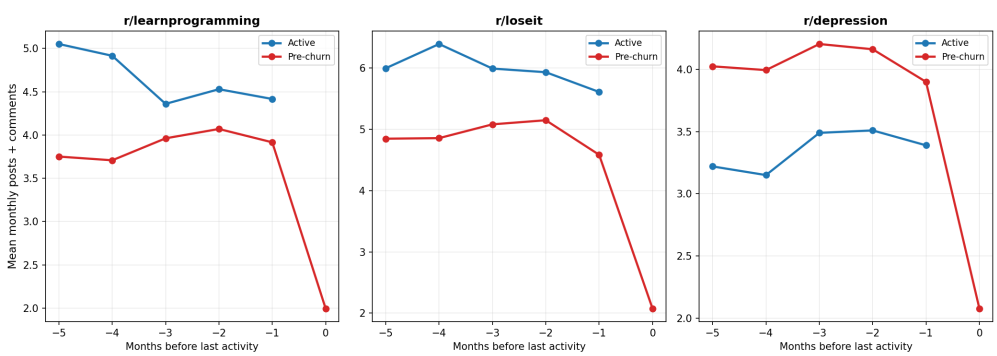


Figure 6: Monthly activity trajectories for active and pre-churn users across three subreddits

Firstly, Figure 6 shows that pre-churn users experience a clear activity decline before disengagement. This decline is much stronger than the change observed among active users. Across the three subreddits, pre-churn users' mean monthly activity dropped by about 47 percent in r/learnprogramming, 57 percent in r/loseit, and 48 percent in r/depression. In contrast, active users stayed relatively stable over the same period. This suggests that recent activity decline is an important signal of upcoming churn. This finding is related to prior Reddit departure research, but adds a more specific temporal pattern: users often reduce activity shortly before they disappear (Trujillo & Cresci, 2023).

Secondly, the decline appears to be relatively sudden rather than gradual. Pre-churn users did not show a steady decrease across the full six-month window. Instead, their activity stayed relatively stable for several months and then dropped sharply near Month -1 and Month 0. This pattern suggests that short-term changes in activity may be more useful than long-term average activity when predicting churn.

However, r/depression also shows a different pattern from the other two communities. In r/depression, pre-churn users were more active than active users for most of the observation window. At Month -5, pre-churn users had a mean monthly activity of 4.03, compared with 3.22 for active users. This suggests that higher activity does not always mean stronger retention in this community. Some users may come to r/depression during a difficult period, participate actively for a short time, and then leave after that period ends (Horta Ribeiro et al., 2021).

Overall, Figure 6 shows that temporal features are important for churn prediction. The strongest signal appears in the final one to two months before disengagement, so slope-based or recency-weighted features may be useful in the modeling stage. The special pattern in r/depression also suggests that the same feature may have different meanings across communities, which supports comparing separate models for the three subreddits.

# 3 Discussion

## 3.1 Conclusion

The initial results provide descriptive evidence for the first part of the research question (RQ1). Both static and temporal analyses show that churned and active users differ in their behavioral patterns. Network features, especially out-degree, in-degree, and k-core, show larger differences between the two groups than NLP features. This suggests that structural embeddedness in the reply network may be a clearer signal of long-term retention than language use alone. The temporal analysis also shows that pre-churn users experience a sharp activity decline in the one to two months before their last observed activity, while active users maintain more stable participation. Together, these patterns provide initial evidence that behavioral signals of churn can be detected before users disappear, and that the timing and meaning of these signals vary across different community types.

## 3.2 Limitations

One limitation is the churn labeling method. The BG/NBD model was initially chosen for churn labeling because it estimates each user's posting rate and dropout probability from frequency, recency, and total observation time, which is better suited to irregular posting behavior than a simple fixed window (Fader et al., 2005). However, the model failed to converge on all three subreddits, most likely because the wide range of T values across subreddits spanning over a decade produced numerical instability in the likelihood optimizer. Multiple configurations were tested, including different penalization coefficients, time unit rescaling, randomization, and alternative starting values, but none achieved convergence.

As a result, churn is defined using the holdout period directly. A user is classified as churned if they made no posts or comments during the six-month holdout window (July 1 to December 31, 2025). This approach is standard in non-contractual churn research (Shahabikargar et al., 2026) and does not require any model fitting.

However, this definition may misclassify users who are genuinely active but post infrequently. A user who posts only a few times per year could have no activity during a six-month window without having permanently left the community. The BG/NBD model would have accounted for this by estimating each user's individual posting rhythm, but that adjustment is not feasible here.

A second limitation is severe class imbalance. Churn rates are very high in all three communities: 97.7 percent in r/learnprogramming, 96.5 percent in r/loseit, and 98.2 percent in r/depression, which means that the active group is much smaller than the churned group. This creates a challenge for the next modeling stage, because a classifier may predict almost all users as churned and still achieve high accuracy. To address this issue, the modeling stage will need to use methods such as class weighting, resampling, or evaluation metrics beyond accuracy.

## 3.3 Future Work

The next step is to train and compare prediction models using the features constructed in the previous stages. I will compare three feature sets: NLP features only, network features only, and a combined NLP plus network feature set. For each feature set, I will train logistic regression and random forest models separately for each subreddit.

Because the data are highly imbalanced, SMOTE oversampling or class-weighted methods will be applied to the training data. Model performance will be evaluated using AUC-ROC and F1-score, rather than accuracy alone. Finally, SHAP values will be used for the combined model to identify which features contribute most to churn prediction. This will help explain whether language features, network features, or both are most useful across different Reddit communities.

# References


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